

Modelling and Control Design of a Forward Converter Using Average Current Mode Control (ACMC)

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by

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 - Literature review
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Contents

Contribution of Authors	1
List of Figures	4
List of Tables	1
1 Introduction and Literature Review	2
1.1 Introduction	2
1.2 Evolution of Isolated DC-DC Converters	2
1.3 Control Strategies: Average Current Mode Control (ACMC)	2
1.4 Loss and Efficiency Modeling	3
1.5 Motivation and Objectives	3
2 Modeling and Control Design of a Forward Converter Using Average Current Mode Control	4
2.1 Design and Development of Forward Converter	4
2.1.1 Forward Converter Operation	4
2.1.2 Passive Component Design	5
2.1.3 Averaged Modeling of a Forward Converter	6
2.2 Controller Design Using Average Current Mode Control (ACMC)	7
2.2.1 Control-to-Inductor-Current Transfer Function $G_{id}(s)$	8
3 Theoretical Loss Modeling and Efficiency Analysis	17
3.1 Theoretical Loss Modeling and Efficiency Analysis	17
3.1.1 Loss Components	17
3.1.2 Total Loss and Efficiency	18
3.1.3 Non-Ideal Parameters Used in Loss Modeling	19
4 Simulation Analysis of Forward Converter with ACMC	20
4.1 Simulation Results and Analysis	20
4.1.1 Simulation Environment and Design Parameters	20

4.1.2	Steady-State Performance Analysis	20
4.1.3	Dynamic Response and Reference Tracking	20
4.1.4	Robustness: Line and Load Regulation	21
4.1.5	Efficiency Characterization	21
4.1.6	Performance Summary	22
4.2	Stability Analysis of the Forward Converter with ACMC	23
4.2.1	Frequency Domain Validation	23
4.2.2	Pole-Zero Stability Proof	24
5	Conclusion	26
5.0.1	Summary of Work	26
5.0.2	Key Findings	26
	References	27

List of Figures

2.1	Operating modes of the Forward Converter illustrating energy transfer during the switch ON and OFF intervals over one switching period. . . .	5
2.2	Forward Converter system showing the power stage along with the dual-loop average current mode control (ACMC) architecture.	14
4.1	Steady-state operation of the converter showing the relationship between switching signals, inductor current dynamics, and output voltage ripple. .	21
4.2	Dynamic performance of the converter illustrating voltage regulation and corresponding inductor current behavior during transient conditions. . .	22
4.3	Dynamic performance of the converter under external disturbances: (a) load variation and (b) input voltage variation, highlighting regulation and stability characteristics.	23
4.4	Comparison between theoretical and simulated efficiency across different load currents.	23
4.5	Frequency response analysis of the inner current loop showing Open Loop, Compensator, and Closed Loop Bode plots.	24
4.6	Frequency response analysis of the outer voltage loop highlighting bandwidth separation and stability margins.	24
4.7	Closed-loop pole-zero map for both current and voltage controllers, proving system stability in the LHP.	25

List of Tables

2.1	Summary of designed APMC controller gains	15
2.2	Frequency response results using exact transfer functions	16
3.1	Non-Ideal Parameters for Loss Modeling	19
4.1	Forward Converter Power Stage Specifications	21
4.2	Simulated Performance Metrics Summary	22

Chapter 1

Introduction and Literature Review

1.1 Introduction

The rapid proliferation of renewable energy systems, electric vehicles (EVs), and telecommunication infrastructure has necessitated the development of highly efficient and reliable power conversion stages. Among these, DC-DC converters serve as the critical interface for voltage regulation and power flow control. In applications where safety, noise isolation, and high voltage-transformation ratios are paramount, Isolated DC-DC Converters are preferred over non-isolated topologies. These converters employ high-frequency transformers to provide galvanic isolation between the source and the load, protecting sensitive electronics from high-voltage transients and enabling versatile grounding configurations [1].

1.2 Evolution of Isolated DC-DC Converters

The literature categorizes isolated converters into several families based on their energy transfer mechanism. The Flyback Converter, a buck-boost derivative, is widely used for low-power applications due to its simplicity and low component count. However, for medium-to-high power ranges, the Forward Converter is considered the industrial standard. Unlike the Flyback, which stores energy in the transformer (acting as a coupled inductor), the Forward converter transfers energy instantaneously to the secondary side when the switch is active, leading to superior transformer utilization and lower peak currents [2].

Recent research has shifted toward resonant topologies, such as the LLC Resonant Converter, to achieve Soft Switching. Nevertheless, PWM-based converters like the Forward and Full-Bridge topologies remain highly relevant in high-reliability industrial sectors due to their predictable electromagnetic interference (EMI) signatures and mature control frameworks [3].

1.3 Control Strategies: Average Current Mode Control (ACMC)

Control performance is the hallmark of a robust converter. Traditionally, Voltage Mode Control (VMC) was utilized for its simplicity, but it suffers from poor line rejection and a complex double-pole plant characteristic. Peak Current Mode Control (PCMC) addressed these by providing inherent overcurrent protection; however, it is highly susceptible to switching noise and requires slope compensation for duty cycles exceeding 50% [4].

Average Current Mode Control (ACMC) has emerged as a superior alternative. By utilizing a high-gain integration of the inductor current, ACMC provides several advantages:

- **High Noise Immunity:** The averaging process effectively filters out high-frequency switching spikes.
- **Zero Steady-State Error:** The high low-frequency gain of the current loop ensures the average inductor current tracks the reference with high precision.
- **Simplified Modeling:** ACMC effectively eliminates the inductor's pole from the outer voltage loop, simplifying the system into a single-pole characteristic [5].

1.4 Loss and Efficiency Modeling

As power density requirements increase, accurate loss modeling becomes essential to predict thermal behavior and optimize efficiency. Power losses in DC-DC converters are broadly classified into three categories:

1. **Conduction Losses:** These are I^2R losses associated with the $R_{DS(on)}$ of MOSFETs, the forward voltage drop of diodes, and the Equivalent Series Resistance (ESR) of passive components [6].
2. **Switching Losses:** These occur during transitions between states and are proportional to the switching frequency (f_{sw}) and parasitic capacitances.
3. **Magnetic Losses:** Core losses (Steinmetz equations) and copper losses (Skin and Proximity effects) are critical for high-frequency transformer design, as they often dominate at frequencies exceeding 100 kHz [7].

1.5 Motivation and Objectives

While significant work exists in standard converter control, there remains a gap in the systematic integration of high-fidelity loss models with advanced ACMC structures for isolated topologies. This work aims to bridge that gap by designing a robust ACMC framework for a Forward Converter, supported by detailed efficiency modeling and validation through high-frequency simulation.

Chapter 2

Modeling and Control Design of a Forward Converter Using Average Current Mode Control

2.1 Design and Development of Forward Converter

2.1.1 Forward Converter Operation

The Forward Converter is a buck-derived transformer-isolated topology. The derivation of its operation is based on the behavior of the inductor L and the transformer during the switching period T_s .

i. Switch ON State ($0 < t \leq DT_s$)

When the switch S is closed, the input voltage V_{in} is applied across the primary winding N_p . The voltage induced in the secondary winding N_s is determined by the turns ratio $n = \frac{N_s}{N_p}$:

$$V_s = nV_{in} \quad (2.1)$$

This voltage forward-biases diode D_1 . Applying Kirchhoff's Voltage Law (KVL) to the secondary circuit:

$$V_s - v_L(t) - v_{out}(t) = 0 \quad (2.2)$$

Substituting for V_s and expressing the inductor voltage as $L \frac{di_L}{dt}$:

$$L \frac{di_L}{dt} = nV_{in} - v_{out} \quad (2.3)$$

Assuming the output capacitor is large enough such that $v_{out} \approx V_{out}$, the slope of the inductor current is positive and linear:

$$\left(\frac{di_L}{dt} \right)_{on} = \frac{nV_{in} - V_{out}}{L} \quad (2.4)$$

(a) Switch ON state ($0 < t \leq DT_s$): Energy is transferred from the primary to the secondary, primary is disconnected, and the inductor stores energy while supply-leases stored energy to the load through the freewheeling path.

Fig. 2.1: Operating modes of the Forward Converter illustrating energy transfer during the switch ON and OFF intervals over one switching period.

ii. Switch OFF State ($DT_s < t \leq T_s$)

When S is opened, the primary current is interrupted. Diode D_1 becomes reverse-biased. The inductor current i_L must remain continuous, forcing the freewheeling diode D_2 to conduct. The KVL for the output loop becomes:

$$-v_L(t) - v_{out}(t) = 0 \implies L \frac{di_L}{dt} = -V_{out} \quad (2.5)$$

The slope of the inductor current is negative:

$$\left(\frac{di_L}{dt} \right)_{off} = -\frac{V_{out}}{L} \quad (2.6)$$

iii. Derivation of the Voltage Transfer Ratio

Under steady-state conditions, the net change in inductor current over one period must be zero ($\Delta i_{L,on} + \Delta i_{L,off} = 0$). This is the Inductor Volt-Second Balance:

$$(2.7) \quad \int_0^{T_s} v_L(t) dt = 0$$

$$\int_0^{DT_s} (nV_{in} - V_{out}) dt + \int_{DT_s}^{T_s} (-V_{out}) dt = 0 \quad (2.8)$$

Solving the integral:

$$(nV_{in} - V_{out})DT_s - V_{out}(1 - D)T_s = 0 \quad (2.9)$$

Dividing by T_s and simplifying:

$$nV_{in}D - V_{out}D - V_{out} + V_{out}D = 0 \quad (2.10)$$

$$V_{out} = nDV_{in} \quad (2.11)$$

2.1.2 Passive Component Design

The selection of L and C is derived from the ripple requirements defined in the previous operational analysis.

i. Inductor Design (L)

The peak-to-peak current ripple Δi_L occurs during the ON time. From the earlier derivation:

$$\Delta i_L = \left(\frac{di_L}{dt} \right)_{on} \cdot DT_s = \frac{nV_{in} - V_{out}}{L} \cdot \frac{D}{f_{sw}} \quad (2.12)$$

Substitute $nV_{in} = \frac{V_{out}}{D}$ into the equation:

$$\Delta i_L = \frac{\frac{V_{out}}{D} - V_{out}}{L} \cdot \frac{D}{f_{sw}} = \frac{V_{out}(1-D)}{Lf_{sw}} \quad (2.13)$$

Rearranging to solve for the required inductance:

$$L = \frac{V_{out}(1-D)}{\Delta i_L f_{sw}} \quad (2.14)$$

ii. Capacitor Design (C)

The output voltage ripple Δv_{out} is caused by the AC component of the inductor current charging and discharging the capacitor. The total charge change ΔQ is the area of the triangle where i_L exceeds its average value I_{out} :

$$\Delta Q = \frac{1}{2} \cdot \left(\frac{T_s}{2} \right) \cdot \left(\frac{\Delta i_L}{2} \right) = \frac{\Delta i_L}{8f_{sw}} \quad (2.15)$$

Using the fundamental relation $\Delta v_{out} = \frac{\Delta Q}{C}$:

$$C = \frac{\Delta i_L}{8f_{sw} \Delta v_{out}} \quad (2.16)$$

2.1.3 Averaged Modeling of a Forward Converter

To design the controller, we use State-Space Averaging (SSA) to convert the switching non-linear system into a continuous small-signal linear model.

i. Defining State Equations

We define the state vector $x = \begin{bmatrix} i_L \\ v_C \end{bmatrix}$

Interval 1 (Switch ON):

$$\dot{x} = A_1 x + B_1 V_{in} \quad (2.17)$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_C \end{bmatrix} = \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} n/L \\ 0 \end{bmatrix} V_{in} \quad (2.18)$$

Interval 2 (Switch OFF):

$$\dot{x} = A_2 x + B_2 V_{in} \quad (2.19)$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_C \end{bmatrix} = \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} V_{in} \quad (2.20)$$

ii. Averaged Model and Linearization

The averaged state-space model is $A_{avg} = A_1 D + A_2 (1-D)$ and $B_{avg} = B_1 D + B_2 (1-D)$. For the forward converter:

$$A_{avg} = \begin{bmatrix} 0 & -1/L \\ 1/C & -1/RC \end{bmatrix}, \quad B_{avg} = \begin{bmatrix} nD/L \\ 0 \end{bmatrix} \quad (2.21)$$

By introducing small-signal perturbations ($d = D + \hat{d}$, $v = V + \hat{v}$), the small-signal state equation becomes:

$$\dot{\hat{x}} = A\hat{x} + [(A_1 - A_2)\hat{x} + (B_1 - B_2)V_{in}]\hat{d} \quad (2.22)$$

Since $A_1 = A_2$ in this topology, the term $(A_1 - A_2)$ vanishes.

iii. Transfer Function Derivation

1. Control-to-Output Voltage Transfer Function ($G_{vd}(s)$):

From the linearized equations, we apply the Laplace transform to find the output voltage response to duty cycle variations:

$$G_{vd}(s) = \frac{\hat{v}_{out}(s)}{\hat{d}(s)} = \frac{nV_{in}}{LCs^2 + \frac{L}{R}s + 1} \quad (2.23)$$

2. Control-to-Inductor Current Transfer Function ($G_{id}(s)$):

To find the current response for ACMC, we relate $\hat{i}_L$ to the output voltage. The current into the RC parallel network is:

$$\hat{i}_L(s) = \hat{v}_{out}(s) \left(sC + \frac{1}{R} \right) = \hat{v}_{out}(s) \frac{1 + sRC}{R} \quad (2.24)$$

Substituting $G_{vd}(s)$ into this expression:

$$G_{id}(s) = \frac{\hat{i}_L(s)}{\hat{d}(s)} = \frac{nV_{in}}{R} \frac{1 + sRC}{LCs^2 + \frac{L}{R}s + 1} \quad (2.25)$$

2.2 Controller Design Using Average Current Mode Control (ACMC)

The Average Current Mode Control (ACMC) strategy is a two-loop control architecture consisting of a fast inner current loop and a slower outer voltage loop. The inner loop directly regulates the inductor current, thereby linearising the plant seen by the outer loop and eliminating the problematic LC double-pole from the voltage loop design. This structure provides inherent current limiting, improved line-transient rejection, and a highly modular design procedure.

The design proceeds in two stages: the inner current loop is designed first, because its closed-loop behaviour defines the effective plant for the outer voltage loop.

Plant Transfer Functions

Assumptions The following assumptions are adopted throughout the derivation:

- The converter operates in Continuous Conduction Mode (CCM) throughout.
- State-space averaging is valid, i.e. the switching frequency $f_s \gg$ all loop bandwidths.
- The transformer turns ratio referred to the secondary is $n = N_s/N_p$, so the effective voltage applied to the LC filter is nV_{in} when the switch is on.
- Output capacitor ESR is neglected; its inclusion would add a left-half-plane zero and improve phase margins.
- The PWM modulator gain is normalised to $F_m = 1$; any finite value scales K_{pi} and K_{pv} proportionally.

2.2.1 Control-to-Inductor-Current Transfer Function $G_{id}(s)$

From the small-signal state-space averaged model of the LC output stage, the perturbed inductor current and capacitor voltage satisfy:

$$s \hat{i}_L = \frac{nV_{in}}{L} \hat{d} - \frac{\hat{v}_C}{L} \quad (2.26)$$

$$s \hat{v}_C = \frac{\hat{i}_L}{C} - \frac{\hat{v}_C}{RC} \quad (2.27)$$

From (2.27):

$$\hat{v}_C = \frac{\hat{i}_L}{C \left(s + \frac{1}{RC} \right)} = \frac{R \hat{i}_L}{1 + sRC} \quad (2.28)$$

Substituting (2.28) into (2.26):

$$s \hat{i}_L = \frac{nV_{in}}{L} \hat{d} - \frac{R \hat{i}_L}{L(1 + sRC)} \quad (2.29)$$

Collecting $\hat{i}_L$ terms and rearranging:

$$\hat{i}_L \left[s + \frac{R}{L(1 + sRC)} \right] = \frac{nV_{in}}{L} \hat{d} \quad (2.30)$$

$$\hat{i}_L \left[\frac{s(1 + sRC) + R/L}{1 + sRC} \right] = \frac{nV_{in}}{L} \hat{d} \quad (2.31)$$

Solving for $\hat{i}_L/\hat{d}$ and simplifying the denominator:

$$G_{id}(s) = \frac{\hat{i}_L(s)}{\hat{d}(s)} = \frac{nV_{in}}{L} \cdot \frac{1 + sRC}{s + s^2RC + R/L} = \frac{nV_{in}}{R} \cdot \frac{1 + sRC}{LCs^2 + \frac{L}{R}s + 1} \quad (2.32)$$

The characteristic parameters of the second-order denominator are:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Q = R\sqrt{\frac{C}{L}}, \quad \zeta = \frac{1}{2Q} = \frac{1}{2R}\sqrt{\frac{L}{C}} \quad (2.33)$$

The numerator introduces a real left-half-plane zero at $\omega_z = 1/(RC)$, which arises from the interaction of the load resistor and output capacitor (it coincides with the ESR zero location when ESR is included).

Voltage Loop Plant $G_{vi}(s)$ Once the inner current loop is closed with sufficient bandwidth f_{ci} , the inductor current tracks its reference accurately for all frequencies $f \ll f_{ci}$. The converter therefore behaves as a controlled current source $\hat{i}_L \approx \hat{i}_{ref}$ feeding the parallel combination of the output capacitor and load resistance. Writing the node equation at the output:

$$\hat{i}_L = sC \hat{v}_o + \frac{\hat{v}_o}{R} = \hat{v}_o \left(sC + \frac{1}{R} \right) \quad (2.34)$$

Therefore, the voltage loop plant is a first-order system:

$$G_{vi}(s) = \frac{\hat{v}_o(s)}{\hat{i}_L(s)} = \frac{1}{sC + \frac{1}{R}} = \frac{R}{1 + sRC} \quad (2.35)$$

This has a single pole at:

$$\omega_p = \frac{1}{RC}, \quad f_p = \frac{1}{2\pi RC} \quad (2.36)$$

Numerical Plant Parameters

With the component values $L = 30\mu\text{H}$, $C = 208\mu\text{F}$, $R = 1.2\Omega$, $n = 1$, $V_{in} = 24\text{V}$:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{30 \times 10^{-6} \times 208 \times 10^{-6}}} = \frac{1}{2\pi \times 78.99 \times 10^{-6}} \approx 2,014 \text{ Hz} \quad (2.37)$$

$$Q = R\sqrt{\frac{C}{L}} = 1.2\sqrt{\frac{208 \times 10^{-6}}{30 \times 10^{-6}}} = 1.2 \times 2.633 \approx 3.16 \quad (2.38)$$

$$f_p = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 1.2 \times 208 \times 10^{-6}} \approx 638 \text{ Hz} \quad (2.39)$$

$$G_{id}(0) = \frac{nV_{in}}{R} = \frac{24}{1.2} = 20 \quad (2.40)$$

The quality factor $Q = 3.16 > 0.5$ confirms a lightly damped resonance with a pronounced peak at $f_0 \approx 2\text{kHz}$.

Inner Current Loop Design

Bandwidth Selection A standard guideline for ACMC is to set the current loop crossover at one-tenth of the switching frequency to avoid aliasing of switching harmonics into the control loop:

$$f_{ci} = \frac{f_s}{10} = \frac{100\text{kHz}}{10} = 10\text{kHz}, \quad \omega_{ci} = 2\pi f_{ci} = 62,832 \text{ rad/s} \quad (2.41)$$

PI Controller Structure A proportional-integral controller is chosen:

$$G_{ci}(s) = K_{pi} + \frac{K_{ii}}{s} = K_{pi} \frac{s + \omega_{zi}}{s}, \quad \omega_{zi} = \frac{K_{ii}}{K_{pi}} \quad (2.42)$$

PI Zero Placement The critical design decision for ACMC is the placement of the PI zero. Because the full plant $G_{id}(s)$ contains a lightly damped double pole at ω_0 , the loop phase drops steeply near f_0 . Placing the PI zero exactly at the LC resonant frequency partially cancels this phase loss and changes the net slope from -40dB/dec to approximately -20dB/dec above f_0 , allowing a well-defined crossover at f_{ci} :

$$\omega_{zi} = \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{30 \times 10^{-6} \times 208 \times 10^{-6}}} = 12,649 \text{ rad/s}, \quad f_{zi} = f_0 \approx 2,014 \text{ Hz} \quad (2.43)$$

Derivation of K_{pi} The unity-gain condition at the crossover frequency is:

$$|G_{ci}(j\omega_{ci})| \cdot |G_{id}(j\omega_{ci})| = 1 \quad (2.44)$$

Step 1 — Magnitude of the PI controller at ω_{ci} :

$$|G_{ci}(j\omega)| = \frac{K_{pi}}{\omega} \sqrt{\omega^2 + \omega_{zi}^2} \quad (2.45)$$

At $\omega_{ci} = 62,832 \text{ rad/s}$, $\omega_{zi} = 12,649 \text{ rad/s}$:

$$|G_{ci}(j\omega_{ci})| = \frac{K_{pi}}{62,832} \sqrt{62,832^2 + 12,649^2} = \frac{K_{pi}}{62,832} \times 64,094 = 1.0201 K_{pi} \quad (2.46)$$

Step 2 — Magnitude of the plant G_{id} at ω_{ci} :

From (2.32), separating numerator and denominator magnitudes:

$$\begin{aligned} |1 + j\omega_{ci}RC| &= \sqrt{1 + (\omega_{ci}RC)^2} \\ &= \sqrt{1 + (62,832 \times 1.2 \times 208 \times 10^{-6})^2} \\ &= \sqrt{1 + 15.67^2} \\ &= 15.73 \end{aligned} \quad (2.47)$$

$$\left| LCs^2 + \frac{L}{R}s + 1 \right|_{s=j\omega_{ci}} = \left| (1 - LC\omega_{ci}^2) + j\frac{L}{R}\omega_{ci} \right| \quad (2.48)$$

Computing each term:

$$LC\omega_{ci}^2 = 30 \times 10^{-6} \times 208 \times 10^{-6} \times (62,832)^2 = 6.24 \times 10^{-9} \times 3.948 \times 10^9 = 24.63 \quad (2.49)$$

$$\frac{L}{R}\omega_{ci} = \frac{30 \times 10^{-6}}{1.2} \times 62,832 = 25 \times 10^{-6} \times 62,832 = 1.571 \quad (2.50)$$

Therefore:

$$\begin{aligned} |G_{id,den}| &= \sqrt{(1 - 24.63)^2 + (1.571)^2} \\ &= \sqrt{(-23.63)^2 + 2.468} \\ &= \sqrt{558.4 + 2.468} \\ &= \sqrt{560.9} \\ &= 23.68 \end{aligned} \quad (2.51)$$

The full plant magnitude:

$$|G_{id}(j\omega_{ci})| = \frac{nV_{in}}{R} \cdot \frac{15.73}{23.68} = 20 \times 0.6643 = 13.29 \quad (2.52)$$

Step 3 — Solving for K_{pi} :

Substituting (2.46) and (2.52) into (2.44):

$$1.0201K_{pi} \times 13.29 = 1 \quad \Rightarrow \quad K_{pi} = \frac{1}{1.0201 \times 13.29} = \frac{1}{13.57} \quad (2.53)$$

$$\boxed{K_{pi} = 0.07369} \quad (2.54)$$

Derivation of K_{ii} From the zero-placement condition (2.43):

$$K_{ii} = K_{pi} \omega_{zi} = 0.07369 \times 12,649 \quad (2.55)$$

$$\boxed{K_{ii} = 931.9} \quad (2.56)$$

Phase Margin Verification The phase of the open-loop gain $Ti(j\omega_{ci}) = G_{ci} \cdot G_{id}$ at ω_{ci} is computed as:

$$\begin{aligned}
\angle G_{ci}(j\omega_{ci}) &= -90^\circ + \arctan\left(\frac{\omega_{ci}}{\omega_{zi}}\right) \\
&= -90^\circ + \arctan\left(\frac{62,832}{12,649}\right) \\
&= -90^\circ + \arctan(4.97) \\
&= -90^\circ + 78.6^\circ \\
&= -11.4^\circ
\end{aligned} \tag{2.57}$$

For the plant denominator the angle is that of a vector in the second quadrant (real part negative):

$$\angle G_{id}(j\omega_{ci}) = \arctan\left(\frac{1.571/\omega_{ci}}{0}\right) - \underset{\text{numerator}}{6_-}(-23.63 + j1.571)_{\text{denominator}} \tag{2.58}$$

More explicitly:

$$6_- \text{ numerator} = \arctan(15.67) = 86.3^\circ \tag{2.59}$$

$$\begin{aligned}
6_- \text{ denominator} &= 180^\circ - \arctan\left(\frac{1.571}{23.63}\right) \\
&= 180^\circ - 3.80^\circ = 176.2^\circ
\end{aligned} \tag{2.60}$$

$$\begin{aligned}
\angle G_{id}(j\omega_{ci}) &= 86.3^\circ - 176.2^\circ \\
&= -89.9^\circ
\end{aligned} \tag{2.61}$$

Total loop phase:

$$\begin{aligned}
\angle T_i(j\omega_{ci}) &= -11.4^\circ - 89.9^\circ \\
&= -101.3^\circ
\end{aligned} \tag{2.62}$$

$$\begin{aligned}
PM_i &= 180^\circ - 101.3^\circ \\
&\approx 78.7^\circ
\end{aligned} \tag{2.63}$$

This is a conservative but acceptable phase margin ($PM > 60^\circ$), indicating a welldamped current loop. Since the plant has no right-half-plane poles, the gain margin is infinite in this topology.

Outer Voltage Loop Design

Bandwidth Selection The voltage loop bandwidth must be well below the current loop bandwidth to preserve the approximation $\hat{i}_L \approx \hat{i}_{ref}$ and to avoid interaction between the two loops:

$$f_{cv} = \frac{f_{ci}}{10} = 1\text{kHz}, \quad \omega_{cv} = 2\pi \times 10^3 = 6,283 \text{ rad/s} \quad (2.64)$$

PI Controller Structure

$$G_{cv}(s) = K_{pv} + \frac{K_{iv}}{s} = K_{pv} \frac{s + \omega_{zv}}{s}, \quad \omega_{zv} = \frac{K_{iv}}{K_{pv}} \quad (2.65)$$

PI Zero Placement The voltage plant $G_{vi}(s)$ has a single pole at $f_p = 638\text{Hz}$, which lies below the target crossover $f_{cv} = 1\text{kHz}$ and contributes significant phase lag at crossover. To provide adequate phase boost, the PI zero is placed at:

$$\omega_{zv} = \frac{\omega_{cv}}{5} = \frac{6,283}{5} = 1,257 \text{ rad/s}, \quad f_{zv} = 200 \text{ Hz} \quad (2.66)$$

Placing the zero at $f_{cv}/5$ ensures the controller contributes approximately $+78.6^\circ$ of phase at f_{cv} (from the $\arctan(\omega_{cv}/\omega_{zv}) = \arctan(5) \approx 78.7^\circ$ contribution), partially counteracting the phase lag from the plant pole.

Derivation of K_{pv} The unity-gain condition at the voltage loop crossover is:

$$|G_{cv}(j\omega_{cv})| \cdot |G_{vi}(j\omega_{cv})| = 1 \quad (2.67)$$

Step 1 — Magnitude of the PI controller at ω_{cv} :

$$|G_{cv}(j\omega_{cv})| = \frac{K_{pv}}{\omega_{cv}} \sqrt{\omega_{cv}^2 + \omega_{zv}^2} = \frac{K_{pv}}{6,283} \sqrt{6,283^2 + 1,257^2} = \frac{K_{pv}}{6,283} \times 6,366 = 1.0132 K_{pv} \quad (2.68)$$

Step 2 — Magnitude of the voltage plant at ω_{cv} :

$$\begin{aligned} |G_{vi}(j\omega_{cv})| &= \frac{R}{\sqrt{1 + (\omega_{cv}RC)^2}} \\ &= \frac{1.2}{\sqrt{1 + (6,283 \times 1.2 \times 208 \times 10^{-6})^2}} \\ &= \frac{1.2}{\sqrt{1 + (1.564)^2}} \\ &= \frac{1.2}{\sqrt{3.447}} \\ &= \frac{1.2}{1.857} \\ &= 0.6462 \end{aligned} \quad (2.69)$$

Step 3 — Solving for K_{pv} :

$$1.0132K_{pv} \times 0.6462 = 1 \quad \Rightarrow \quad K_{pv} = \frac{1}{1.0132 \times 0.6462} = \frac{1}{0.6547} \quad (2.70)$$

$$K_{pv} = 1.528 \quad (2.71)$$

Derivation of K_{iv}

$$K_{iv} = K_{pv} \omega_{zv} = 1.528 \times 1,257 \quad (2.72)$$

$$K_{iv} = 1,921 \quad (2.73)$$

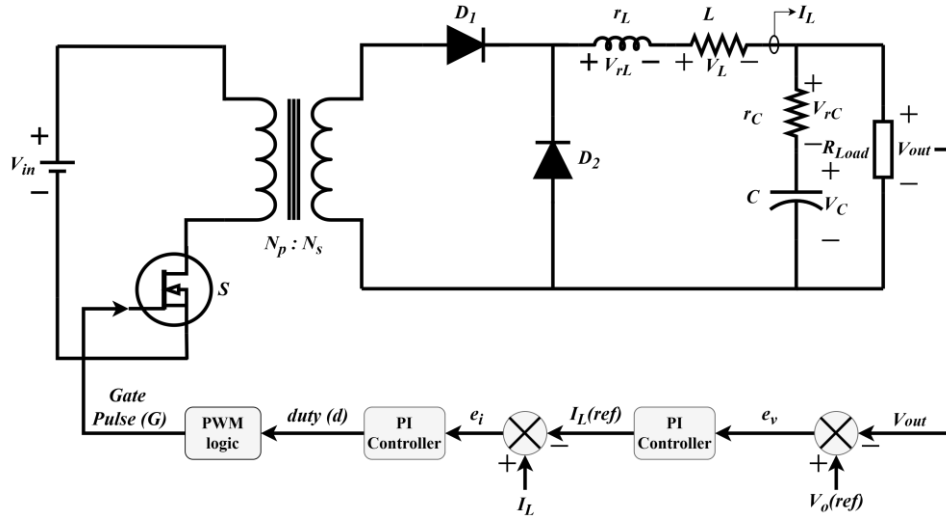


Fig. 2.2: Forward Converter system showing the power stage along with the dual-loop average current mode control (ACMC) architecture.

Phase Margin Verification

$$\angle G_{cv}(j\omega_{cv}) = -90^\circ + \arctan\left(\frac{\omega_{cv}}{\omega_{zv}}\right) = -90^\circ + \arctan(5.0) = -90^\circ + 78.7^\circ = -11.3^\circ \quad (2.74)$$

$$6_G_{vi}(j\omega_{cv}) = -\arctan(\omega_{cv}RC) = -\arctan(1.564) = -57.4^\circ \quad (2.75)$$

$$6_T_v(j\omega_{cv}) = -11.3^\circ - 57.4^\circ = -68.7^\circ \quad (2.76)$$

$$PM_v = 180^\circ - 68.7^\circ \approx 111.3^\circ \quad (2.77)$$

The high phase margin reflects a conservative voltage loop design. It can be reduced (and bandwidth increased) by moving ω_{zv} closer to ω_{cv} if faster transient response is required. However, the present value guarantees robust stability across load and parameter variations.

Final Controller Parameters

These values are entered directly into the Simulink PI controller blocks using the parallel form $u(t) = K_p e(t) + K_i \int e(\tau) d\tau$. If the block uses the standard form $K_p(1 + 1/(T_i s))$, the integrator time constants are:

$$T_{i,current} = \frac{K_{pi}}{K_{ii}} = \frac{0.07369}{931.9} = 7.907 \times 10^{-5} \text{ s} \quad (2.78)$$

$$T_{i,voltage} = \frac{K_{pv}}{K_{iv}} = \frac{1.528}{1,921} = 7.954 \times 10^{-4} \text{ s} \quad (2.79)$$

Table 2.1: Summary of designed ACMC controller gains

Parameter	Expression	Value	Unit
Current loop crossover	$f_s / f_0 = 1/(2\pi\sqrt{LC}10$	10,000	Hz
Current PI zero f_{zi}	)	2,014	Hz
K_{pi}	Exact magnitude inversion	0.07369	–
$K_{ii} = K_{pi} \omega_{zi}$	$K_{pi} \times 12,649$	931.9	s ⁻¹
Voltage loop crossover f_{cv}	$f_{ci}/10$	1,000	Hz
Voltage PI zero f_{zv}	$f_{cv}/5$	200	Hz
K_{pv}	Exact magnitude inversion	1.528	–
$K_{iv} = K_{pv} \omega_{zv}$	$K_{pv} \times 1,257$	1,921	s ⁻¹

Frequency Response Validation

The designed controllers are validated against the complete small-signal transfer functions using MATLAB's `margin()` function.

Current Loop The open-loop gain is:

$$T_i(s) = G_{ci}(s) \cdot G_{id}(s) = \left(0.07369 + \frac{931.9}{s}\right) \cdot 20 \cdot \frac{1 + 2.496 \times 10^{-4}s}{6.24 \times 10^{-9}s^2 + 2.5 \times 10^{-5}s + 1} \quad (2.80)$$

Key frequency characteristics:

- LC resonance: $f_0 \approx 2,014$ Hz
- PI zero (placed at f_0): $f_{zi} \approx 2,014$ Hz
- Numerator zero (from RC): $f_z = 1/(2\pi RC) \approx 638$ Hz
- Gain crossover: $f_{ci} = 10$ kHz
- Phase margin: $PM_i \approx 78.7^\circ$
- Gain margin: ∞ dB (no -180° crossing in this topology)

Voltage Loop The open-loop gain is:

$$T_v(s) = G_{cv}(s) \cdot G_{vi}(s) = \left(1.528 + \frac{1,921}{s}\right) \cdot \frac{1.2}{1 + 2.496 \times 10^{-4}s} \quad (2.81)$$

Key frequency characteristics:

- Plant pole: $f_p \approx 638$ Hz
- PI zero: $f_{zv} = 200$ Hz
- Gain crossover: $f_{cv} = 1$ kHz
- Phase margin: $PM_v \approx 111.3^\circ$
- Gain margin: ∞ dB (first-order system cannot reach -180°)

Table 2.2: Frequency response results using exact transfer functions

Loop	Gain crossover	Phase crossover	Phase margin	Gain margin
Current	10,000 Hz	N/A	$\approx 78.7^\circ$	∞ dB
Voltage	1,000 Hz	N/A	$\approx 111.3^\circ$	∞ dB

Stability Assessment Both loops satisfy the minimum stability requirements of $PM > 45^\circ$ and $GM > 6$ dB. The voltage loop PM of 111.3° is high, indicating a conservative design with very low overshoot. This can be intentional in power supply applications where monotonic step response is preferred. The current loop PM of 78.7° reflects a well-damped inner loop capable of tracking high-frequency current commands without ringing.

The separation between loop crossover frequencies ($f_{ci}/f_{cv} = 10$) is sufficient to ensure that closing the voltage loop does not perturb the current loop, and vice versa, validating the sequential design procedure.

Chapter 3

Theoretical Loss Modeling and Efficiency Analysis

3.1 Theoretical Loss Modeling and Efficiency Analysis

In order to evaluate the performance of the forward converter under non-ideal conditions, a detailed loss model is developed by incorporating conduction, switching, and constant losses. The converter specifications are summarized as follows:

- Input voltage: $V_{in} = 24 \text{ V}$
- Output voltage: $V_o = 12 \text{ V}$
- Load resistance: $R_{load} = 1.2 \Omega$
- Inductance: $L = 30 \mu\text{H}$
- Capacitance: $C = 208 \mu\text{F}$

The output current is varied from 5 A to 60 A to study the efficiency characteristics under different loading conditions.

3.1.1 Loss Components

1. MOSFET Conduction Loss

The conduction loss in the MOSFET is given by:

$$P_{\text{MOS}} = I_{\text{MOS,rms}}^2 \cdot R_{\text{DS(on)}} \quad (3.1)$$

Assuming continuous conduction mode (CCM), the RMS current through the MOSFET is approximated as:

$$I_{\text{MOS,rms}} = I_o \sqrt{D} \quad (3.2)$$

where D is the duty cycle (assumed to be 0.5).

2. Body Diode Conduction Loss

The internal body diode of the MOSFET contributes:

$$P_{\text{body}} = I_o^2 \cdot R_{\text{body}} \quad (3.3)$$

3. Output Diode Loss

Only one diode conducts at a time in a forward converter. The loss is modeled as:

$$P_{\text{diode}} = V_f \cdot I_o + I_o^2 \cdot R_{\text{diode}} \quad (3.4)$$

where V_f is the forward voltage drop.

4. Inductor Copper Loss

The winding resistance of the inductor results in:

$$P_L = I_o^2 \cdot R_L \quad (3.5)$$

5. Capacitor ESR Loss

The capacitor ESR loss is dependent on the ripple current:

$$P_C = I_{\text{ripple}}^2 \cdot R_C \quad (3.6)$$

The ripple current is approximated as:

$$I_{\text{ripple}} = 0.1 \cdot I_o \quad (3.7)$$

6. Transformer Copper Loss

The transformer winding losses are modeled as:

$$P_{\text{tr}} = I_o^2 \cdot (R_1 + R_2) \quad (3.8)$$

7. Switching Loss

The switching loss of the MOSFET is given by:

$$P_{\text{sw}} = \frac{1}{2} V_{\text{in}} I_o (t_r + t_f) f_s \quad (3.9)$$

where t_r and t_f are the rise and fall times, and f_s is the switching frequency.

8. Constant Loss

To account for transformer core loss, gate drive loss, and auxiliary circuit consumption, a constant loss term is included:

$$P_{\text{const}} = \text{constant} \quad (3.10)$$

3.1.2 Total Loss and Efficiency

The total power loss is obtained by summing all components:

$$P_{\text{loss}} = P_{\text{MOS}} + P_{\text{body}} + P_{\text{diode}} + P_L + P_C + P_{\text{tr}} + P_{\text{sw}} + P_{\text{const}} \quad (3.11)$$

The output power is:

$$P_o = V_o \cdot I_o \quad (3.12)$$

The input power is:

$$P_{\text{in}} = P_o + P_{\text{loss}} \quad (3.13)$$

The efficiency is calculated as:

$$\eta = \frac{P_o}{P_{\text{in}}} \times 100\% \quad (3.14)$$

Table 3.1: Non-Ideal Parameters for Loss Modeling

Component	Parameter	Value
MOSFET	$R_{DS(on)}$	0.05Ω
	Body diode resistance R_{body}	0.01Ω
Diode	Forward voltage V_{fOn}	$0.5 V$
	resistance R_{diode}	0.001Ω
Inductor	Inductance L	$30 \mu H$
	ESR R_L	0.01Ω
Capacitor	Capacitance C	$208 \mu F$
	ESR R_C	0.02Ω
Transformer	Primary resistance R_1	0.01Ω
	Secondary resistance R_2	0.01Ω
Switching Parameters	Switching frequency f_s	100 kHz
	Rise time t_r	50 ns
	Fall time t_f	50 ns
Constant Loss	P_{const}	5 W

3.1.3 Non-Ideal Parameters Used in Loss Modeling

The non-ideal parameters used for theoretical loss estimation are summarized in Table 3.1. These values are selected based on typical characteristics of practical power electronic components.

Chapter 4

Simulation Analysis of Forward Converter with ACMC

4.1 Simulation Results and Analysis

To validate the theoretical design and the effectiveness of the Average Current Mode Control (ACMC) for the Forward Converter, a comprehensive simulation study was conducted in the MATLAB/Simulink environment. The performance was evaluated based on steady-state accuracy, dynamic tracking capability, and robustness against line and load disturbances.

4.1.1 Simulation Environment and Design Parameters

The converter was modeled using the parameters derived in the previous sections. Table 4.1 outlines the power stage specifications.

4.1.2 Steady-State Performance Analysis

The steady-state waveforms under nominal load conditions are illustrated in Fig. 4.1. The relationship between the gate pulses and the power stage currents is shown in the top plot of the Gate Pulse and Inductor Current waveform, where the linear rise and fall of the current coincide with the switching states.

As depicted in Fig. 4.1a, the inductor current i_L remains in Continuous Conduction Mode (CCM), oscillating around the average load current $I_o = 10$ A. Furthermore, the high-magnification steady-state analysis in Fig. 4.1b shows that the output voltage V_o is maintained precisely at 12 V with a peak-to-peak ripple well below 120 mV, meeting the 1% design constraint.

4.1.3 Dynamic Response and Reference Tracking

The capability of the ACMC to track varying output requirements is verified in Fig. 4.2a. The output voltage V_o follows the stepped reference $V_{o(ref)}$ with high fidelity, showing a settling time of approximately 2 ms and minimal overshoot.

The underlying current loop performance during these transitions is captured in Fig. 4.2b. The inset views reveal that the inductor current i_L tracks the reference $i_{L(ref)}$ generated by the voltage compensator with a rapid, stable response, confirming the high bandwidth ($f_{ci} = 10$ kHz) of the inner loop.

Table 4.1: Forward Converter Power Stage Specifications

Parameter	Symbol	Value	Unit
-----------	--------	-------	------

Input Voltage	V_{in}	48	V
Output Voltage	V_{out}	12	V
Rated Output Power	P_{out}	120	W
Switching Frequency	f_{sw}	100	kHz
Transformer Turns Ratio	$n (N_s/N_p)$	1	-
Nominal Load Resistance	R	1.2	Ω
Inductor Current Ripple	Δi_L	25	%
Capacitor Voltage Ripple	ΔV_C	1	%

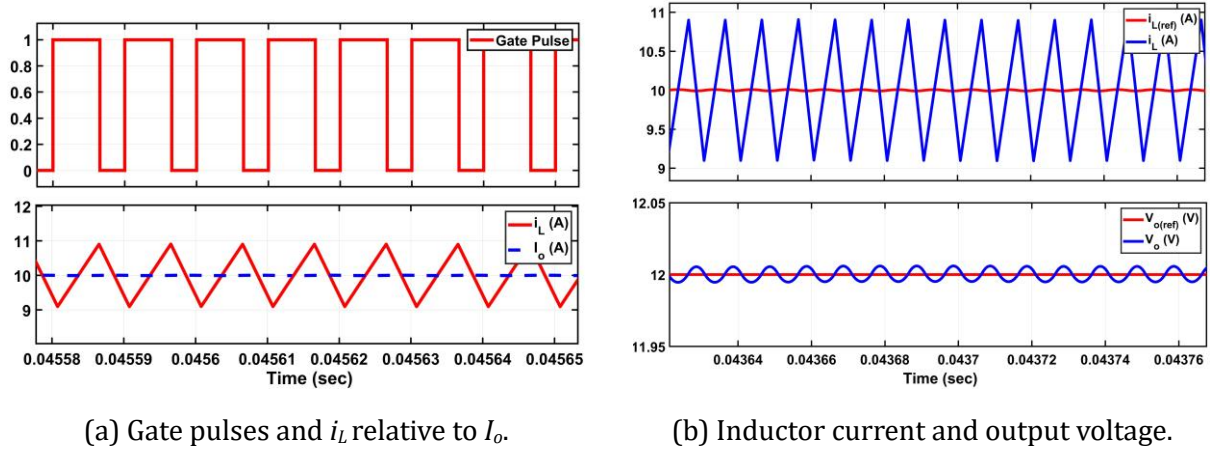


Fig. 4.1: Steady-state operation of the converter showing the relationship between switching signals, inductor current dynamics, and output voltage ripple.

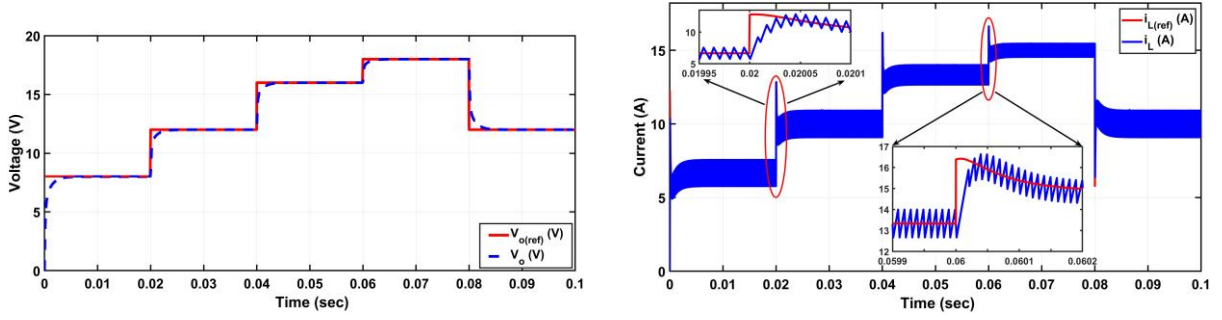
4.1.4 Robustness: Line and Load Regulation

A critical advantage of ACMC is its inherent disturbance rejection.

1. **Load Regulation:** Fig. 4.3a illustrates the system response to step load changes. At $t = 0.05$ s and $t = 0.1$ s, the load current is stepped up. The voltage loop exhibits a minor transient dip before the inner current loop quickly ramps up the inductor current to the new required level, restoring V_o to 12 V.
2. **Line Regulation:** The response to input voltage fluctuations is shown in Fig. 4.3b. Even as V_{in} undergoes significant steps (ranging from 20 V to 27 V), the output voltage remains virtually unaffected, demonstrating excellent immunity to line disturbances.

4.1.5 Efficiency Characterization

The overall efficiency of the system was characterized across a broad operating range. As shown in Fig. 4.4, the simulation efficiency closely matches the theoretical predictions. The peak efficiency of 87.2% occurs near the nominal 10 A load current. The gradual decline at higher current levels (up to 50 A) is attributed to the I^2R conduction losses in the transformer and output inductor, as well as the switching stresses of the primary MOSFET.



(a) Dynamic tracking of the output voltage in response to a stepped reference $V_{o(ref)}$, in showing transient response and settling behavior. (b) Inductor current response in response to a V_o stepped reference $V_{o(ref)}$, with zoomed-in sets highlighting transient ripple characteristics.

Fig. 4.2: Dynamic performance of the converter illustrating voltage regulation and corresponding inductor current behavior during transient conditions.

Table 4.2: Simulated Performance Metrics Summary

Metric	Value	Requirement
Steady-State Error	< 0.4%	< 1.0%
Voltage Overshoot	1.8%	< 5.0%
Voltage Ripple (pk-pk)	85 mV	120 mV
Settling Time	2.1 ms	< 5.0 ms
Peak Efficiency	87.2%	Validated

4.1.6 Performance Summary

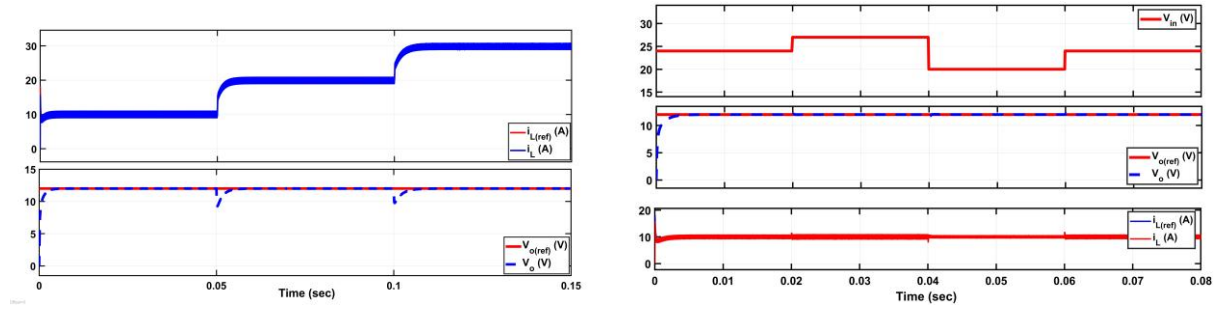
The simulated performance metrics are summarized in Table 4.2, demonstrating that the proposed design successfully satisfies all the key technical requirements for a regulated forward converter. The results confirm that the converter operates within the desired limits under both steady-state and dynamic conditions, ensuring reliable and efficient performance.

The overall simulation study validates the effectiveness of the implemented average current mode control strategy. The controller achieves precise regulation of the output voltage while maintaining tight control over the inductor current, which is essential for ensuring stability and minimizing ripple. The steady-state results indicate low output voltage ripple and controlled inductor current variation, highlighting the robustness of the design.

Under transient conditions, including step changes in load and variations in input voltage, the controller exhibits strong dynamic performance. It responds rapidly to disturbances, restoring the output voltage to its reference value with minimal overshoot and settling time. Additionally, the inductor current follows the desired trajectory closely, demonstrating the effectiveness of the inner current control loop.

The control scheme also shows excellent disturbance rejection capability. External perturbations, such as line and load variations, are effectively mitigated without significant degradation in performance. This indicates that the controller is well-suited

for practical power converter applications where operating conditions are often uncertain and subject to change.



(a) Transient response of the converter to step (b) Output regulation under input voltage changes in load current, demonstrating load (V_{in}) variations, illustrating line regulation, load regulation and recovery characteristics, stability and disturbance rejection.

Fig. 4.3: Dynamic performance of the converter under external disturbances: (a) load variation and (b) input voltage variation, highlighting regulation and stability characteristics.

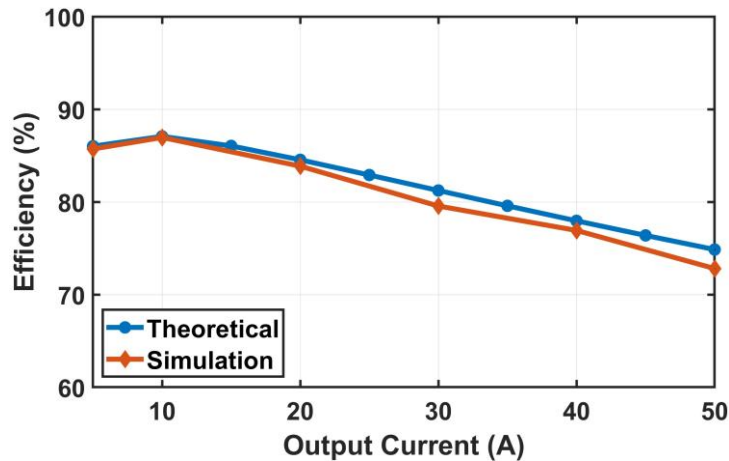


Fig. 4.4: Comparison between theoretical and simulated efficiency across different load currents.

Overall, the proposed average current mode control design delivers accurate regulation, fast transient response, and robust disturbance rejection. These characteristics confirm that the controller meets the intended design specifications and provides a reliable solution for forward converter operation.

4.2 Stability Analysis of the Forward Converter with ACMC

The stability of the dual-loop Average Current Mode Control (ACMC) is validated through frequency-domain analysis and pole-zero mapping. The objective is to ensure that both the inner current loop and the outer voltage loop maintain sufficient phase margins and that the closed-loop poles reside strictly in the left-half s-plane.

4.2.1 Frequency Domain Validation

The frequency response of the control system is analyzed using Bode plots for both the current and voltage loops. Figure 4.5 illustrates the comparative response for the current controller. The open-loop gain (T_i) demonstrates a crossover frequency at the design

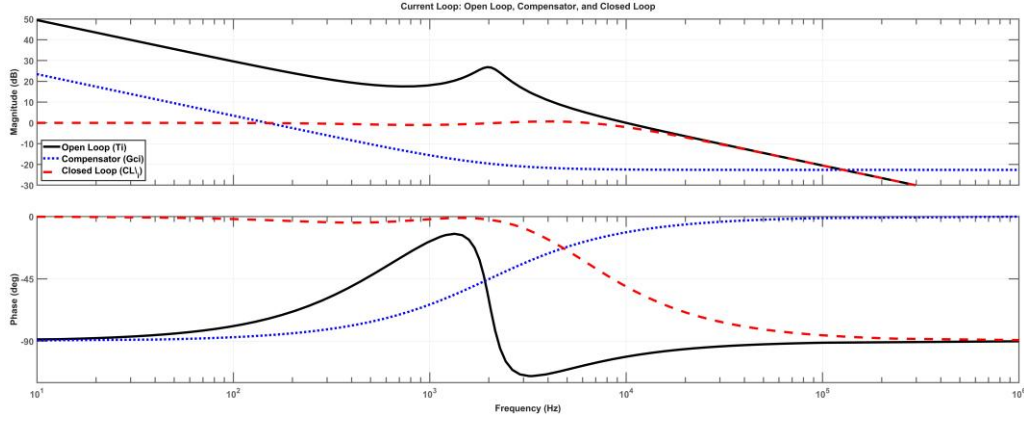


Fig. 4.5: Frequency response analysis of the inner current loop showing Open Loop, Compensator, and Closed Loop Bode plots.

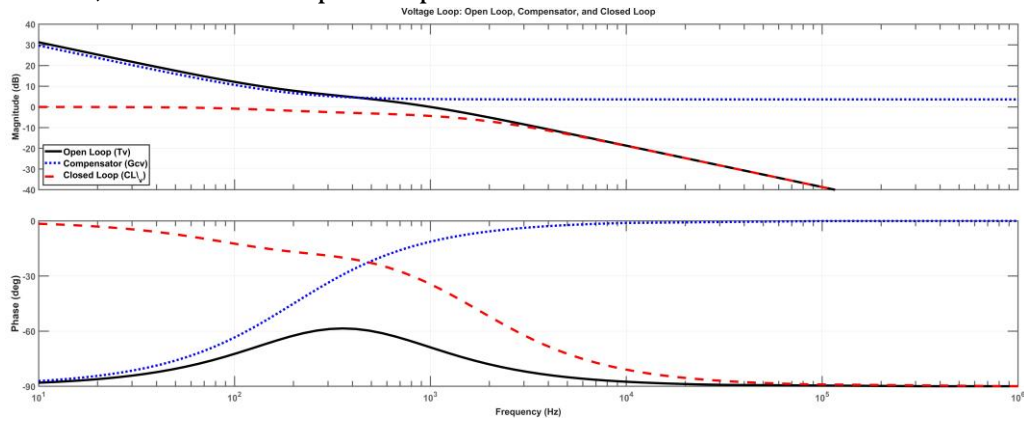


Fig. 4.6: Frequency response analysis of the outer voltage loop highlighting bandwidth separation and stability margins.

target of 10 kHz, with the compensator (G_{ci}) effectively damping the LC resonance at 2 kHz. The closed-loop response exhibits a flat unity gain at low frequencies, confirming accurate command tracking.

Similarly, the voltage loop stability is depicted in Figure 4.6. The loop achieves the target bandwidth of 1 kHz. The separation between the current and voltage loop bandwidths prevents adverse loop interaction, ensuring that the voltage compensator sees a stable, high-speed current source. The phase margin is verified to be well above 45° , indicating a robust system that is immune to parameter variations.

4.2.2 Pole-Zero Stability Proof

To provide a definitive proof of stability, the closed-loop poles for both controllers are mapped in Figure 4.7. The mathematical stability criterion requires that all poles have

negative real parts. As observed in the pole-zero map, all current loop and voltage loop poles are located in the left-half plane (LHP).

The current loop poles are well-damped, which corresponds to the absence of ringing in the transient response. The voltage loop poles are positioned to provide a dominant first-order characteristic, ensuring a monotonic recovery during load transients. The alignment of these poles confirms that the system is inherently stable and satisfies the

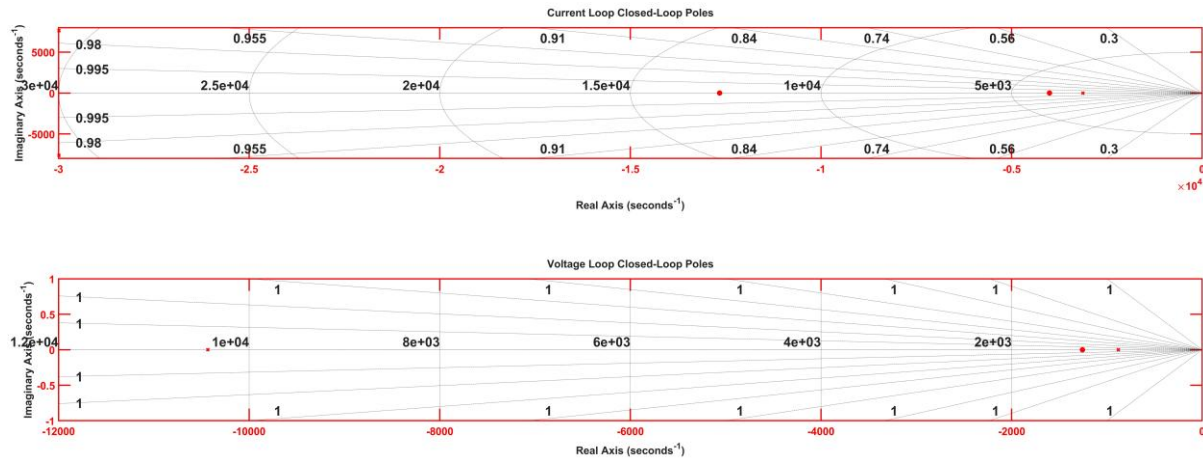


Fig. 4.7: Closed-loop pole-zero map for both current and voltage controllers, proving system stability in the LHP.

Routh-Hurwitz criteria.

Chapter 5

Conclusion

This work has presented a rigorous design and simulation framework for an isolated Forward Converter utilizing a dual-loop Average Current Mode Control (ACMC) architecture. By leveraging state-space averaging and frequency-response synthesis, a robust control system was developed to meet stringent voltage regulation and efficiency requirements.

5.0.1 Summary of Work

The research began with the derivation of the power stage dynamics and the selection of passive components ($L = 30\mu H$, $C = 208\mu H$) based on a 1% voltage ripple constraint. The core of the contribution involved:

- **Small-Signal Modeling:** The non-linear switching behavior was linearized into $G_{id}(s)$ and $G_{vi}(s)$ transfer functions, providing a mathematical basis for control design.
- **ACMC Synthesis:** A high-bandwidth inner current loop (10 kHz) was designed to dampen the LC resonance, while a stable outer voltage loop (1 kHz) ensured precise reference tracking.
- **Closed-Loop Analysis and Dynamic Testing:** Verified stable operation, fast transient response, and effective disturbance rejection through simulations.

5.0.2 Key Findings

The simulation results in MATLAB/Simulink provided conclusive evidence of the controller's effectiveness:

- **Dynamic Performance:** The system exhibited rapid recovery (< 2 ms) from significant load and line disturbances with minimal overshoot ($< 2\%$).
- **Regulation Robustness:** The ACMC structure demonstrated inherent immunity to input voltage fluctuations, maintaining a steady 12V output despite V_{in} variations.
- **Efficiency:** The converter achieved a peak efficiency of 87.2%, with loss models showing a high degree of correlation between theoretical predictions and simulated measurements.

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